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HUNNY GUPTA
9999903256, 9999603256

HT Board Booster

Class X Science

Biology

CBSE Sample Paper Question Bank

Chapter wise Practice

Chapters 7 - 8
Reproduction and Haredity

Session 2026-27

Practice | Perform | Progress

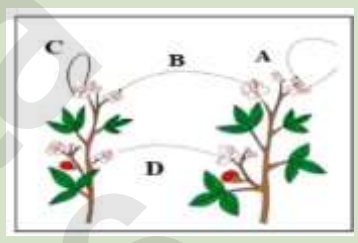
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Your CBSE Partner

Hunny Gupta

Class X Biology
CBSE SQP 2019-2025
Ch- Reproduction

Section - A

Q.1. The diagram shown below depicts pollination. Choose the options that will show a maximum variation in the offspring **(CBSE SQP 2022-23)**



(a) A, B and C
(b) B and D
(c) B, C and D
(d) A and C

Q.2. A farmer wants to grow banana plants genetically similar enough to the plants already available in his field. Which one of the following methods would you suggest for this purpose? **(CBSE SQP 2023-24)**

(a) Regeneration (b) Budding
(c) Vegetative propagation (d) Sexual reproduction

Q.3. **Assertion:** Probability of survival of an organism produced through sexual reproduction is more than that of organism produced through asexual mode. **(CBSE SQP 2023-24)**

Reason: Variations provide advantages to individuals for survival.

Q.4. In snails' individuals can begin life as male and depending on environmental - conditions they can become female as they grow. This is because **(CBSE SQP 2024-25)**

(a) Male snails have dominant genetic makeup.
(b) Female snails have dominant genetic makeup.
(c) Expression of sex chromosomes can change in a snail's life time.
(d) Sex is not genetically determined in snails.

Section – B

Q.5. State the post-fertilisation changes that lead to fruit formation in plants. **(CBSE SQP 2023-24)**

Section – C

Q.6. What is the probability of a girl or a boy being born in a family? Justify your answer. **(CBSE SQP 2023-24)**

Section – E

Q.7. (a) Describe the various steps involved in the process of binary fission with the help of a diagram. **(CBSE SQP 2019-20)**

(b) Why do multicellular organisms use complex way of reproduction?

Hunny Gupta 9999903256

OR

(a) Describe the role of prostate gland, seminal vesicle and testes in the human male Reproductive system.

(b) How is the surgical removal of unwanted pregnancies misused?

(c) Explain the role of oral contraceptive pills in preventing conception.

Q.8. Trace the changes that take place in a flower from gamete formation to fruit formation. **(CBSE SQP 2020-21)**

Q.9. (a) Why is it not possible to reconstruct the whole organism from a fragment in complex multicellular organisms? **(CBSE SQP 2022-23)**

(b) Sexual maturation of reproductive tissues and organs are necessary link for reproduction. Elucidate.

OR

(a) How are variations useful for species if there is drastic alteration in the niches?

(b) Explain how the uterus and placenta provide necessary conditions for proper growth and development of the embryo after implantation?

Q.10. Given below are certain situations. Analyze and describe its possible impact on a person:

(a) Testes of a male boy are not able to descend into scrotum during his embryonic development. **(CBSE SQP 2023-24)**

(b) Vas deferens of a man is plugged.

(c) Prostate and seminal vesicles are not functional.

(d) Egg is not fertilised in a human female.

(e) Placenta does not attach to the uterus optimally.

Q.11. **Attempt either option A or B.** **(CBSE SQP 2025-26)**

A. Puneet wanted to grow banana plants.

(i) Based on your knowledge on plant reproduction should he opt for seeds or any alternate method of reproduction. Justify your answer.

(ii) Offsprings of a banana plant usually show very little variation. What causes variation and are variations good or bad? Justify.

OR

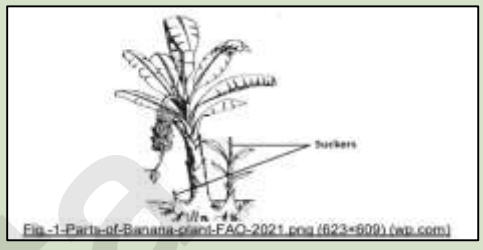
B. Annie was conducting research on the number of fruits produced by watermelon under different conditions. She grew 25 watermelon plants each in both glass house A and B. She introduced pollinators in glass house A only.

(i) What difference will she observe in the number of fruits produced in the two glass houses? Explain with reason.

(ii) List 3 changes that will occur in a flower once it gets fertilized.

Hunny Gupta 9999903256

Q.12. The image below shows a banana plant which is growing with the help of suckers. These suckers are small plant stem outgrowths which can be separated from the main plant and planted separately and they will grow into a new plant subsequently.



(i) Give the name and type of reproduction that is shown in the image Above.

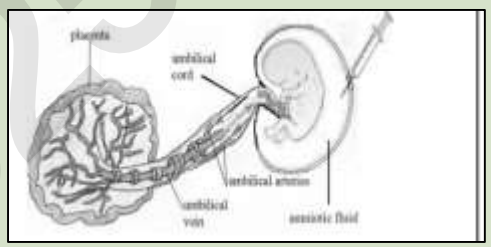
(ii) List two advantages the farmer will have on using the type of reproduction mentioned above.

(iii) The above plant produces male flowers. Explain how this plant will be involved in the process of pollination.

(iv) Why is the offspring of this banana plant not absolutely identical to its parent plant?

OR **(CBSE SQP 2024-25)**

The image below shows a developing fetus in the mother's womb. The developing fetus is connected to the placenta by means of umbilical cord. The Umbilical vein and artery run inside the umbilical cord.



(i) Name two substance that moves through the blood vessels.

(ii) If the placenta has less villi how will it affect the baby's growth?

(iii) Name the region where the embryo develops inside the female body. Explain how this region is adapted for nourishing the baby.

(iv) Some of the fetal cells fall off into the amniotic fluid and can be collected by careful procedure. The cells were screened and found to contain XY chromosome.

(a) What is the sex of the foetus?

(b) How is this prenatal sex determination misused?

Hunny Gupta 9999903256

Class X Biology
CBSE SQP 2019-2025
Ch- Heredity and Its Evolution

Section – A

Q.1. **Assertion (A):** A geneticist crossed a pea plant having violet flowers with a pea plant with white flowers, he got all violet flowers in first generation. **(CBSE SQP 2020-21)**
Reason (R): White colour gene is not passed on to next generation.

Q.2. If a tall pea plant is crossed with a pure dwarf pea plant then, what percentage of F1 and F2 generation respectively will be tall? **(CBSE SQP 2022-23)**
(a) 25%, 25% (b) 50%, 50% (c) 75%,100% (d) 100%, 75%

Q.3. **Assertion (A):** Height in pea plants is controlled by efficiency of enzymes and is thus genetically controlled. **(CBSE SQP 2022-23)**
Reason (R): Cellular DNA is the information source for making proteins in the cell.

Q.4. In snails' individuals can begin life as male and depending on environmental -conditions they can become female as they grow. This is because **(CBSE SQP 2024-25)**
(a) Male snails have dominant genetic makeup.
(b) Female snails have dominant genetic makeup.
(c) Expression of sex chromosomes can change in a snail's life time.
(d) Sex is not genetically determined in snails.

Q.5. **Assertion (A):** Generally, the number of chromosomes in a cell and in a germ 1 cell is not the same in species. **(CBSE SQP 2024-25)**
Reason (R): When two germ cells combine, they restore the normal number of chromosomes in a species.

Q.6. In a cross between black furred rabbit (B) and white furred rabbit (b), all offspring were found to have black fur. What can be inferred about the genetic makeup of the parent rabbits?
(a) BB X bb (b) Bb X Bb (c) Bb X bb (d) bb X bb
(CBSE SQP 2025-26)

Q.7. **Assertion (A):** Tallness of a pea plant is controlled by an enzyme. **(CBSE SQP 2025-26)**
Reason (R): The gene for that enzyme makes proteins which help the plant to be tall.

Section – C

Q.8. In a pea plant, the trait of flowers bearing purple colour (PP) is dominant over white colour (pp). Explain the inheritance pattern of F1 and F2 generations with the help of a cross following the rules of inheritance of traits. State the visible characters of F1and F2 progenies. **(CBSE SQP 2019-20)**

Hunny Gupta 9999903256

Q.9. After self-pollination in pea plants with round, yellow seeds, following types of seeds were obtained by Mendel:

Seeds	Number
Round, yellow	630
Round, green	216
Wrinkled, yellow	202
Wrinkled, green	64

Analyse the result and describe the mechanism of inheritance which explains these results.

OR

In humans, there is a 50% probability of the birth of a boy and 50 % probability that a girl will be born. Justify the statement on the basis of the mechanism of sex-determination in human beings. **(CBSE SQP 2020-21)**

Q.10. (a) In a family of four individuals, the father possessed long ears and the mother possessed short ears. If the parents had pure dominant and recessive traits respectively, then calculate the ratio of genetic makeup of F2 generation. Show a suitable cross. **(CBSE SQP 2024-25)**

(b) If father had short ears and the mother had long ears, explain what effect it will have on the ratio of genetic makeup in F2 generation.

Q.11. In a genetic experiment, plants with pure round green seeds (RRyy) were crossed with plants with wrinkled yellow seeds (rrYY). **(CBSE SQP 2025-26)**

(i) Show the gametes formed when F1 was self-pollinated.

(ii) A total of 144 seeds were produced which developed into saplings. Show the ratio in which these traits are independently inherited in these 144 sapling.

Section – E

Q.12. Pooja has green eyes while her parents and brother have black eyes. Pooja's husband Ravi has black eyes while his mother has green eyes and father has black eyes.

(a) On the basis of the above given information, is the green eye colour a dominant or recessive trait? Justify your answer. **(CBSE SQP 2022-23)**

(b) What is the possible genetic makeup of Pooja's brother's eye colour?

(c) What is the probability that the offspring of Pooja and Ravi will have green eyes? Also, show the inheritance of eye colour in the offspring with the help of a suitable cross.

OR

(d) 50% of the offspring of Pooja's brother are green eyed. With help of cross show how this is possible.

Hunny Gupta 9999303256

Q.13. Figures (a) to (d) given below represent the type of ear lobes present in a family consisting of 2 children - Rahul, Nisha and their parents.



Excited by his observation of different types of ear lobes present in his family, Rahul conducted a survey of the type of ear lobes found {Figure (e) and (f)} in his classmates. He found two types of ear lobes in his classmates as per the frequency given below:

Sex	**Free**	**Attached**
Male	36	14
Female	31	19

On the basis of above data answer the following questions.

(a) Which of the two characteristics 'free ear lobe' or 'attached ear lobe' appears to be dominant in this case? Why?

(b) Is the inheritance of the free ear lobe linked with sex of the individual? Give reason for your answer.

(c) What type of ear lobe is present in father, mother, Rahul and his sister Nisha? Write the genetic constitution of each of these family members which explains the inheritance of this character in this family?

(Gene for Free ear lobe is represented by F and gene for attached ear lobe is represented by f for writing the genetic constitution).

OR

Suresh's parents have attached ear lobes. What type of ear lobe can be seen in Suresh and his sister Siya? Explain by giving the genetic composition of all.

Hunny Gupta 9999903256